T–S–837a

January 10, 1955
SUPERSEDING
Fed. Spec. T–S–837
29 January 1951

FEDERAL SPECIFICATION

SWABS, DECK (WITH OR WITHOUT HANDLES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers deck swabs primarily used for wet-cleaning decks of ships and other floorings.

1.2 Classification.

1.2.1 *Types.*—Deck swabs covered by this specification shall be of the following types, as specified:

Type I.—With attached handle (see fig. 1).

Type II.—With attachable handle and as specified in 1.2.1.1 (see fig. 2).

1.2.1.1 *Items (type II only).*—Type II deck swabs shall be furnished as specified in the invitation for bids, contract, or order, as follows:

Item 1.—Swab head, plug, and handle.

Item 2.—Swab head and plug, without handle.

Item 3.—Handle only.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 The following Federal Specifications, of the issues in effect on the date of invitation for bids, form a part of this specification:

T–R–571—Rope; Cotton.

T–R–675—Rope, Yarn and Twine; Hemp.

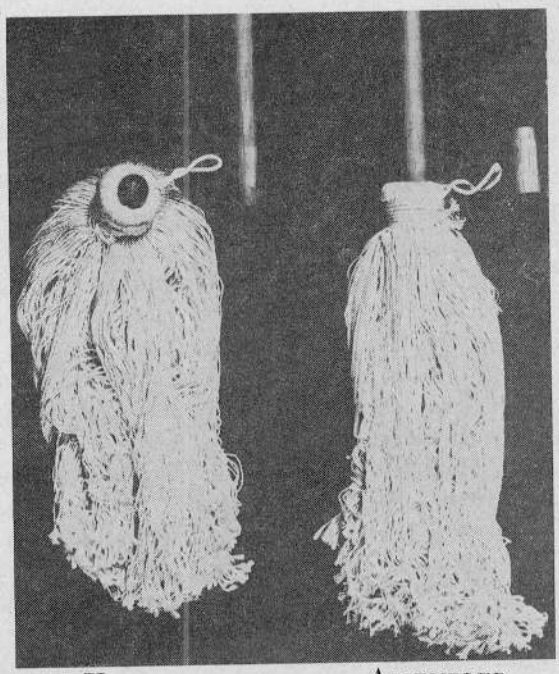
T–T–881—Twine; Cotton, Seine.

CCC–T–191—Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of



FIGURE 1.—TYPE I DECK SWAB.



UNASSEMBLED ASSEMBLED

FIGURE 2.—TYPE II DECK SWAB.

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Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Specifications and other publication applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Wood (for handles).*—The wood for handles shall be of either of the following groups of hardwood or softwood:

Hardwood—White ash (tough), beech, birch, hickory, elm, maple, pecan, white oak, and red oak.

Softwood—Douglas fir, and southern yellow pine.

3.1.2 *Swab yarn.*—The cotton swab yarn shall be at least 8-ply bleached or unbleached cotton yarn not less than 600 yards to the pound, having a breaking strength of not less than 15 pounds.

3.1.3 *Wrapping twine.*—The wrapping twine shall be cotton seine twine in accordance with Federal Specification T–T–881, except that it shall be 60-thread having a minimum breaking strength of 108 pounds and a yardage of approximately 100 yards per pound. An alternate for cotton seine twine is medium tarred marline in accordance with Federal Specification T–R–675.

3.2 Handles.—The wood handles shall be properly seasoned, free from knots, shake, wane, worm holes, heart centers, rot, dote and brash. Cross-grain and spiral-grain with a slope of less than 1 inch in 15 inches will not be acceptable (see 6.2).

3.2.1 *Length and diameter.*—The handles shall be 48½ plus or minus ½ inches in length, straight, rounded at both ends, and of the following diameters:

Hardwood—$\frac{15}{16}$ inch minimum.
$1\frac{5}{16}$ inches maximum.
Softwood —1 inch minimum.
$1\frac{5}{16}$ inches maximum.

3.2.2 *Finish.*—The surface of hardwood and softwood handles shall be sanded smooth and coated with a clear lacquer, varnish, or similar surface preparation to maintain the natural wood color and prevent the absorption of water.

3.2.2.1 The following types of wood handles shall have two applications of the coating material: Douglas fir, southern yellow pine, elm, white oak, and red oak.

3.2.2.2 The following types of wood handles shall have one application of coating material: White ash (tough), birch, beech, hickory, maple, and pecan.

3.3 Swab head.—Swab heads shall consist of not less than 1½ pounds of cotton yarn in strands not less than 38 inches in length attached to swab handle (type I) or plug (type II) by means of wrapping twine (see 3.1.3). The finished swab head shall measure not less than 17 inches in length.

3.4 Type I swab, with attached handle.

3.4.1 *Method of securing swab to handle.*—Two methods for securing the swab to the handle are described below. Unless otherwise specified, method No. 1 shall be used; method No. 2 shall be used only when specified.

3.4.1.1 *Method No. 1.*—One and one-fourth inches from the swab end, the handle shall be notched down to a diameter of ⅛ to $\frac{3}{16}$ inch less than its nominal diameter and tapered up approximately 2 inches to the full diameter of the handle. A $\frac{7}{32}$-inch hole shall pass through the center of and perpendicular to the axis of the handle at a distance of approximately 2¼ inches from the swab end. (This handle can also be used for type II swab.) The swab shall be secured to the handle by laying the center of the swab over the end of the handle and making six tight turns of wrapping twine around the swab and handle, and passing the twine through the $\frac{7}{32}$-inch hole in the handle. The ends of the swab toward the handle shall then be doubled over on the six turns of twine

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and another tight wrapping of eight turns of twine made neatly around the outside of the swab.

3.4.1.2 *Method No. 2.*—Two 2-inch dowels shall be inserted and nailed in 7/32-inch holes passing through the center of the handle at right angles to each other, perpendicular to its main axis, approximately 2½ inches and 2 inches, respectively, from the swab end. (This handle cannot be used with type II swabs.) The swab shall be secured to the handle by laying the center of the swab over the end of the handle and making six tight turns of medium tarred marline or seine twine around the handle above the dowels. The swab shall be held by a 1-inch lacquer-coated saddle-head nail. The end of the swab toward the handle shall then be doubled over the six turns of marline or seine twine and another tight wrapping of eight turns of the same twine made neatly around the handle below the dowels.

3.5 Type II swab, with attachable handle.

3.5.1 Each type II swab head shall be properly fitted on a wooden plug (see 3.5.2 and 3.5.3) with wrapping twine conforming to the requirements of 3.1.3 or with 3/16-inch nominal diameter cotton rope in accordance with Federal Specification T–R–571.

3.5.2 *Plug.*—The plug shall be of wood, shall be 3 inches long, and shall be tapered, having a small diameter of 1 1/16 inches and a large diameter of 1¼ inches.

3.5.3 *Assembly.*—The center of the swab shall be placed over the large end of the plug and six tight turns of wrapping twine shall be made around the swab and the plug. The ends of the swab yarn shall be doubled over the six turns and another tight wrapping of eight turns shall be neatly made around the outside of the swab as shown in figure 1. The outside diameter of the swab head (that portion which receives the end of a swab handle of the design described in 3.4.1.1) measured across the wrapping twine shall be not less than 2 3/8 inches.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Two swabs shall be taken from each lot of 100 swabs or fraction thereof.

4.1.1 When the following twines, yarn, or rope are used for the deck swabs, samples of each one used and not less than the following specified lengths shall be furnished by the contractor for test purposes:

Cotton rope—23 feet.
Marline twine—123 feet.
Seine twine—323 feet.

4.1.2 *Swab yarn.*—Not less than 60 feet of swab yarn shall be furnished by the contractor for test purposes.

4.2 Inspection.—Sample swabs shall be subjected to surface inspection by the inspector to determine compliance with the requirements of this specification.

4.3 Tests.—Tests shall be conducted in accordance with the applicable methods described in Federal Specification T–R–675, T–R–571, T–T–881, and the following methods described in CCC–T–191:

Tests	Method No.
Standard conditions	Section 4.
Identification of hemp	1220.
Identification of cotton	1200.
Strength, breaking (twine, rope)	4102.
Strength, breaking (yarn)	4100.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, the items shall be bundled in units of 10. The bundles shall be tightly bound at both ends with not less than four turns of marline. Type I swabs shall be bundled with five swab heads at each end of the bundle.

5.2 Packing.—The swabs shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.—Unless otherwise specified, shipping containers shall be marked with the name and type of the commodity, the item number for type II as defined by the contract or order under which the shipment is made, the quantity contained therein, the name of the contractor, and the number of the contract or order.

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6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein and should specify the type and item (where applicable) required (see 1.2, 3.4.1, 5.1, 5.3, 7.2.3, and 7.2.4).

6.2 Handles.—A cross grain is one in which the grain of the wood does not run parallel with the center of the axis of the handle, and means that for a handle 1 inch in diameter, the grain must not cross from one side to the other in a distance less than that specified. It is to be noted that white ash of a tough texture is required for hardwood handles and not the cabinet-type ash which is unsuitable for wood handles.

6.3 Transportation description.—The transportation description applicable to this item is:

Deck swabs:

Mop heads, cotton, not otherwise indexed by name.
Carload minimum weight—any quantity.
Mop heads, cotton, without metal.
Truckload minimum weight—24,000 pounds.

Handles:

Handles, wooden, not otherwise indexed by name.
Carload minimum weight—36,000 pounds.
Truckload minimum weight—36,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army and Navy.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

CCC–B–811—Burlap; Jute.
CCC–O–721—Osnaburg; Cotton.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Sampling.*

7.2.2.1 *Lot.*—For the purpose of sampling, a lot shall consist of not more than 5,000 swabs offered for delivery at one time.

7.2.2.2 *Sampling procedure.*—The inspector shall take samples from each lot in accordance with the sampling plan shown in table I, except as specified. The samples, as far as practicable, shall be representative of the lot. When the material offered under a contract or order is of sufficient quantity so that 10 successive lots of the manufacturer's product offered have been accepted under the normal sampling plan (see table I), the reduced sampling plan shown in table II may be applied to the sampling of the balance of material offered under the contract or order. If any lot is rejected under the reduced plan (see table II), resumption of sampling under the normal plan (see table I) shall be thereupon applied.

TABLE I.—*Normal sampling plan*

Number of swabs in inspection lot	Number of swabs to be inspected	Defective swabs	
		Acceptance number	Rejection number
65 and under	7	0	1
66 to 500	25	1	2
501 to 800	35	2	3
801 to 1, 300	50	3	4
1, 301 and over	75	4	5

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TABLE II.—*Reduced sampling plan*

Number of swabs in inspection lot	Number of swabs to be inspected	Defective swabs	
		Acceptance number	Rejection number
65 and under	5	0	1
66 to 500	5	1	2
501 to 800	7	1	2
801 to 1, 300	10	2	3
1, 301 and over	15	2	3

7.2.2.3 *Rejection.*—If any sample fails to conform to any of the requirements of this specification, the lot shall be rejected. Rejected lots may be resubmitted for inspection after the manufacturer has inspected the lot 100 percent and removed or repaired all defective swabs.

7.2.3 *Packaging.*—Unless otherwise specified, deck swabs shall be bundled in units of 10, alternating swab ends. The bundles shall be tightly bound at both ends with not less than four turns of marline.

7.2.4 *Packing.*

7.2.4.1 *For domestic shipment.*—Unless otherwise specified, deck swabs, bundled as described in 7.2.3, shall have the swab ends wrapped in kraft paper, or its equivalent, having a minimum basis weight of 60 pounds. The wrapping shall be secured with suitable twine.

7.2.4.2 *For overseas shipment.*—Unless otherwise specified, deck swabs packed as described in 7.2.4.1 shall be packed for overseas shipment by bundling four units together. Each bundle shall be wrapped in osnaburg or burlap conforming to Federal Specification CCC–O–721 or CCC–B–811, respectively, and securely tied with twine.

7.2.5 *Marking.*—In addition to any special marking required by the contract or order, each bundle shall be marked in accordance with Military Standard MIL–STD–129.

7.2.6 *Ordering data.*—Procurement documents should specify the kind of overseas wrapper required, and whether the deck swabs are to be packed for domestic or overseas shipment (see 7.2.4.1 and 7.2.4.2).

MILITARY INTEREST:
Army—T Q
Navy—Sh

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